

Common Network Ports — Research Notes

Ethical Hacking Task 02 | White Band Associates Internship

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This document provides a concise reference on eleven commonly encountered network ports. For each port, the transport protocol, functional purpose, and key security risks are described. Understanding these ports is essential during the enumeration phase of ethical hacking, as they represent the primary attack surface of most networked systems.

Port 20 / 21	FTP — File Transfer Protocol	TCP
Protocol:	TCP	
Purpose:	FTP is one of the oldest protocols for transferring files between a client and a server over a network. Port 21 is the control channel (for commands and authentication) while port 20 is the data transfer channel used in active mode. FTP is still used in legacy systems, web hosting environments, and certain industrial applications.	
Security Risks:	• Transmits usernames, passwords, and file data in plaintext — easily captured via packet sniffing.	
	• Anonymous FTP login is sometimes enabled, allowing unauthenticated access to files.	
	• Vulnerable to brute-force attacks, bounce attacks, and man-in-the-middle (MitM) interception.	
	• Recommendation: Replace with SFTP (SSH File Transfer Protocol) or FTPS (FTP over TLS).	

Port 22	SSH — Secure Shell	TCP
Protocol:	TCP	
Purpose:	SSH provides a secure, encrypted channel for remote command-line access, file transfers (via SCP/SFTP), and port forwarding. It replaced the insecure Telnet protocol and is the standard method for remote administration of Linux/Unix servers.	
Security Risks:	• Exposed SSH ports are heavily targeted by automated brute-force and credential stuffing attacks.	
	• Weak or default passwords significantly increase the risk of unauthorized access.	
	• Outdated SSH versions may be vulnerable to known exploits (e.g., OpenSSH CVEs).	
	• Recommendation: Use key-based authentication, disable root login, change the default port, and use fail2ban.	

Port 23	Telnet	TCP
Protocol:	TCP	
Purpose:	Telnet was one of the earliest remote access protocols, allowing users to log into remote systems over a network and execute commands. It operates completely without encryption, transmitting all data including credentials in cleartext.	
Security Risks:	• All communication, including passwords, is sent in plaintext and can be captured by anyone on the network.	
	• Highly vulnerable to MitM attacks, session hijacking, and eavesdropping.	
	• Many legacy industrial devices (routers, PLCs) still have Telnet enabled by default.	
	• Recommendation: Completely disable Telnet and use SSH for all remote access.	

Port 25	SMTP — Simple Mail Transfer Protocol	TCP
Protocol:	TCP	
Purpose:	SMTP is the standard protocol used to send and relay email messages between mail servers. Email clients use SMTP to submit outgoing messages to their mail server, and mail servers use SMTP to relay messages to recipient servers.	
Security Risks:	• Open SMTP relays can be exploited by spammers to send mass unsolicited emails through your server.	
	• SMTP without encryption (TLS/STARTTLS) exposes email content and credentials in transit.	
	• Email spoofing is possible if SPF, DKIM, and DMARC records are not properly configured.	
	• Recommendation: Enforce STARTTLS, configure SPF/DKIM/DMARC, and restrict relay access.	

Port 53	DNS — Domain Name System	TCP / UDP
Protocol:	TCP / UDP	
Purpose:	DNS translates human-readable domain names (e.g., www.google.com) into IP addresses that computers use to communicate. UDP port 53 is used for standard queries; TCP port 53 is used for zone transfers and large responses.	
Security Risks:	• DNS zone transfers (AXFR) can expose the entire internal DNS structure of an organization to attackers.	
	• DNS cache poisoning/spoofing can redirect users to malicious sites without their knowledge.	
	• DNS amplification attacks are a common DDoS vector using open resolvers.	

	• DNS tunneling can be used to exfiltrate data from networks that block other outbound traffic.
	• Recommendation: Restrict zone transfers, use DNSSEC, and disable recursive queries on public resolvers.

Port 80	HTTP — HyperText Transfer Protocol	TCP
Protocol:	TCP	
Purpose:	HTTP is the foundation of data communication on the World Wide Web. Web browsers use HTTP to request and receive web pages, images, and other resources from web servers. Port 80 is the default unencrypted web port.	
Security Risks:	• All data, including form inputs and cookies, is transmitted in cleartext and can be intercepted.	
	• Vulnerable to man-in-the-middle attacks, SSL stripping, and session hijacking.	
	• Web servers on port 80 are common targets for directory traversal, SQL injection, and XSS attacks.	
	• Recommendation: Redirect all HTTP traffic to HTTPS (port 443) using SSL/TLS.	

Port 110	POP3 — Post Office Protocol v3	TCP
Protocol:	TCP	
Purpose:	POP3 allows email clients to retrieve messages from a mail server. Unlike IMAP, POP3 typically downloads emails to the local device and deletes them from the server. It is designed for single-device offline email access.	
Security Risks:	• POP3 without SSL/TLS transmits email content and login credentials in plaintext.	
	• Downloaded emails are removed from the server, making multi-device access difficult.	
	• Vulnerable to credential brute-forcing and packet sniffing on unencrypted connections.	
	• Recommendation: Use POP3S (port 995, POP3 over SSL) or switch to IMAP with TLS.	

Port 143	IMAP — Internet Message Access Protocol	TCP
Protocol:	TCP	
Purpose:	IMAP allows email clients to access and manage messages stored on a mail server without downloading them locally. It supports multiple devices accessing the same mailbox simultaneously and is the preferred protocol for modern email clients.	
Security Risks:	• Unencrypted IMAP (port 143) exposes email content and credentials in transit.	
	• Vulnerable to brute-force attacks on login credentials.	

	• IMAP servers can be targeted for email harvesting if improperly secured.
	• Recommendation: Use IMAPS (port 993, IMAP over SSL/TLS) with strong password policies.

Port 443	HTTPS — HTTP Secure	TCP
Protocol:	TCP	
Purpose:	HTTPS is the secure version of HTTP, using TLS (Transport Layer Security) to encrypt all communication between a web browser and server. It ensures confidentiality, data integrity, and server authentication. All modern websites should use HTTPS exclusively.	
Security Risks:	• Expired or self-signed SSL/TLS certificates can be exploited for MitM attacks.	
	• Weak cipher suites and outdated TLS versions (TLS 1.0/1.1) can be compromised.	
	• Even with HTTPS, the web application layer may still be vulnerable to injection attacks, XSS, etc.	
	• HTTPS does not protect against vulnerabilities in the web application itself.	
	• Recommendation: Use TLS 1.2 or 1.3, strong ciphers, and keep certificates updated via automation (e.g., Let's Encrypt).	

Port 445	SMB — Server Message Block	TCP
Protocol:	TCP	
Purpose:	SMB is a network file-sharing protocol used primarily in Windows environments for sharing files, printers, and other resources across a network. It also supports inter-process communication (IPC) between networked systems.	
Security Risks:	• SMB is the vector for the EternalBlue exploit used in WannaCry and NotPetya ransomware attacks.	
	• SMBv1 is critically vulnerable and should never be enabled.	
	• Exposed SMB on the internet is one of the most dangerous misconfigurations possible.	
	• Lateral movement within networks is commonly achieved through SMB exploitation.	
	• Recommendation: Disable SMBv1, block port 445 at the firewall for internet-facing systems, and keep patches current.	

Port 3389	RDP — Remote Desktop Protocol	TCP
Protocol:	TCP	
Purpose:	RDP is a Microsoft protocol that allows users to remotely control a Windows desktop over a network connection. It provides a full graphical interface to the remote machine and is widely used for IT administration and remote work.	

Security Risks:	• RDP exposed to the internet is a top entry point for ransomware attacks and brute-force attempts.
	• BlueKeep (CVE-2019-0708) and DejaBlue are critical RDP vulnerabilities enabling remote code execution.
	• Credential stuffing and brute-force attacks against RDP logins are extremely common.
	• RDP sessions can be hijacked if not properly secured.
	• Recommendation: Never expose RDP directly to the internet; use VPN, Network Level Authentication (NLA), and change the default port.